Liv Functional

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PCOS & Insulin Reversal.

Metabolic Restoration.

Your symptoms are not random. They are the downstream consequence of one hormone running your biology unchecked. This protocol fixes the root, not the list.

Reham

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Before You Begin

> ⚠️ **Medical Disclaimer**
>
> This protocol is educational. I am not your doctor. I am not a dietitian. I am a certified holistic nutrition coach and a T1D who has tested everything in these pages on my own blood sugar before recommending it to anyone else. Nothing here replaces a diagnosis, a prescription, or a conversation with your physician. If you are on medication — metformin, spironolactone, hormonal contraceptives — do not stop or adjust your medication without medical supervision. Run all supplement additions by your prescribing physician if you are on any medications.

What this protocol is : a structured, research-informed intervention targeting the metabolic root of PCOS — hyperinsulinemia — through nutrition, movement, supplementation, sleep, and stress regulation. Sixty days is enough time to move the needle on fasting insulin, free testosterone, and inflammatory markers. It is not enough to undo a decade of damage. Manage that expectation going in.

What This Protocol Covers

PCOS is not a reproductive problem. It is a metabolic problem with reproductive consequences. The research is clear: 70–80% of women with PCOS have measurable insulin resistance — even those with a "normal" BMI. This is where the protocol starts.

> **What You Will Address**
>
> - Chronic hyperinsulinemia
> - Androgen excess (high free testosterone)
> - Systemic inflammation (elevated CRP, IL-6)
> - HPA axis dysregulation (cortisol → DHEA-S)

- Anovulation driven by LH:FSH imbalance
- SHBG suppression
- Gut microbiome disruption
- Poor mitochondrial efficiency

What This Protocol Is Not For

- Active fertility treatment cycles
- Women currently on clomiphene or letrozole
- Eating disorder history without professional support
- Unmanaged thyroid disease (address thyroid first)
- Anyone wanting a quick fix or macro counting approach

How to Use This Protocol

This is a sequenced intervention. The phases are not interchangeable. Phase 1 creates the metabolic baseline that Phase 2 builds on. Skipping ahead is the most common mistake.

Phase	Days	Primary Target	The Work
Phase 1	1–20	Blood sugar stabilization	Eliminate glucose spikes, establish eating structure, begin foundational supplements
Phase 2	21–40	Hormonal recalibration	Lower androgens, reduce inflammation, support the HPA axis, introduce cycle tracking
Phase 3	41–60	Metabolic resilience	Advanced insulin strategies, stress metabolism, sleep architecture, long-term integration

CGM Recommendation

If you can access a continuous glucose monitor (CGM) — Libre, Dexcom, or equivalent —

> use it for at least the first 30 days. PCOS-driven insulin resistance is often invisible on standard fasting glucose. A CGM reveals the glucose spikes and extended recovery periods that a single lab number misses entirely. It is the single highest-leverage data tool available to you.

What to Gather Before Day 1

Lab Work (Baseline)

- Fasting insulin (target: <5 µIU/mL)
- Fasting glucose + HbA1c
- HOMA-IR (calculated)
- Free + total testosterone
- SHBG
- LH:FSH ratio
- DHEA-S
- hs-CRP
- Vitamin D3 (25-OH)
- Ferritin, full iron panel
- Full thyroid panel (TSH, Free T3, Free T4)

Tools & Supplies

- Food scale (non-negotiable for first 2 weeks)
- CGM if accessible
- Blood pressure cuff
- Glucose meter (backup)
- This workbook printed or digital

- Cycle tracking app (Clue, Natural Cycles)
- Sleep tracker (Oura, Apple Watch)

The Root Cause: Why Insulin Runs PCOS

You cannot treat PCOS symptoms without understanding the mechanism. Most conventional approaches treat the end of the chain — the irregular periods, the acne, the weight — without touching what's driving it upstream.

> "High insulin is not a side effect of PCOS. In most cases, it is the cause."

The Mechanism in Plain Language

When you eat — especially refined carbohydrates and sugar — your pancreas releases insulin to move glucose out of the blood and into cells. In insulin-resistant tissue, cells don't respond properly. The pancreas compensates by releasing more insulin. This chronic high-insulin state — hyperinsulinemia — has cascading consequences specifically for women:

- Ovarian androgen overproduction. Insulin directly stimulates the ovarian theca cells to produce more testosterone and androstenedione. This is the primary driver of elevated androgens in PCOS — not a broken ovary.
- SHBG suppression. High insulin suppresses the liver's production of sex hormone-binding globulin (SHBG) — the protein that binds free testosterone and renders it inactive. Less SHBG means more free testosterone circulating, driving acne, hirsutism, and hair loss.
- LH hypersecretion. Hyperinsulinemia disrupts the hypothalamic-pituitary axis, amplifying LH pulses. An elevated LH:FSH ratio prevents normal follicle maturation — the mechanism behind anovulation and irregular cycles.

- Systemic inflammation. Insulin resistance activates inflammatory pathways (NF-κB, IL-6, TNF-α). Chronic low-grade inflammation further impairs insulin receptor function — a self-reinforcing cycle.
- Adrenal amplification. High cortisol (from chronic stress, poor sleep, undereating) drives adrenal androgen production independently of the ovaries — explaining why stress makes PCOS symptoms spike.

> **HOMA-IR: Your Most Important Starting Number**
>
> HOMA-IR = (Fasting Insulin × Fasting Glucose) ÷ 405 Optimal: <1.5 | Insulin Resistant: >2.0 | Severe: >3.0 Standard labs will not calculate this for you. Do the math yourself. Most women with PCOS are above 2.0 even when their fasting glucose is "normal."

The PCOS Phenotype You're Working With

PCOS is not one condition. It has four recognized phenotypes. The nutrition and supplement approach in this protocol is calibrated for the metabolic phenotype (high androgens + insulin resistance ± polycystic ovaries). If you have the lean/normal-weight phenotype, all protocols apply — your insulin resistance is just less visible on standard panels. If you have the inflammatory phenotype (normal androgens, irregular cycles, elevated CRP), Phase 1 and Phase 3 are your highest leverage areas.

Blood Sugar First.

Week One · Days 1–7

The Elimination

Remove what is driving insulin up before adding anything new

The Non-Negotiable Remove List

This week is not about perfection. It is about removing the primary insulin drivers so your baseline can reveal itself. These are not debatable for the first 20 days.

Remove Completely	Why It Matters for PCOS
All added sugar (including honey, agave, "natural" sweeteners)	Fructose specifically drives hepatic insulin resistance and suppresses SHBG production
Refined grains (bread, pasta, rice, crackers)	Rapid glucose spikes → hyperinsulinemia → direct androgen stimulus
Seed oils (canola, sunflower, soybean, corn oil)	High omega-6 content drives prostaglandin-mediated inflammation; amplifies insulin resistance
Alcohol	Impairs liver glucose regulation, disrupts sleep architecture, elevates cortisol
Dairy milk and flavored yogurts	High insulinotropic index regardless of glycemic load; research-linked to IGF-1 elevation in PCOS
Processed "diet" or "low-fat" foods	Contain hidden sugars, maltodextrin, or starches that spike insulin without satiety
Caffeine after 12pm	Cortisol half-life; afternoon caffeine blunts melatonin onset, disrupting overnight insulin sensitivity restoration

The Eating Structure: Week 1

> **Core Rule: Protein + Fat Before Carbohydrate at Every Meal**

> The order in which you eat macronutrients significantly affects your post-meal glucose response. Eating protein and fat first (even 10 minutes before carbohydrates) reduces the glucose peak by up to 38% in published research. This is the cheapest, most accessible tool in this protocol. Use it at every meal, starting today.

Daily Eating Template

Meal	Timing	Composition
Meal 1 (Break Fast)	Within 90 min of waking — no later	35–40g protein · 15–25g fat · <15g net carbs · No fruit juice, no cereal, no toast
Meal 2 (Midday)	4–6 hours after Meal 1	30–40g protein · liberal vegetables · moderate fat · 20–30g net carbs max
Meal 3 (Evening)	3–4 hours before bed	30–35g protein · non-starchy vegetables · healthy fat · <20g net carbs

> ⚠️ **On Intermittent Fasting**
>
> Extended fasting (16:8 or longer) is not recommended during Phase 1. For women with PCOS — especially those with elevated cortisol, adrenal involvement, or history of restriction — skipping meals raises cortisol, which directly stimulates adrenal androgen production. A structured 12-hour overnight fast (finish dinner by 7pm, break fast by 7am) is sufficient and safe. We revisit this in Phase 3.

Movement: Week 1

- Post-meal walks (10–15 minutes) after every meal. This is not optional. Walking post-meal is the fastest, most evidence-backed tool for blunting the glucose spike by

contracting muscle GLUT4 transporters independently of insulin. Start with the evening walk if only doing one.

- No high-intensity cardio this week. Elevated cortisol from intense cardio will counteract the insulin-stabilizing work. Walk, stretch, swim at conversational pace.

Supplement Stack — Week 1 Foundations

Introduce these in Week 1. Do not add Phase 2 supplements yet.

Supplement	Dose	Timing	Why
Myo-Inositol	2,000 mg twice daily	With breakfast · With dinner	Restores insulin receptor signaling in ovarian tissue; improves FSH sensitivity; directly lowers LH:FSH ratio in meta-analyses. The most evidence-backed supplement in PCOS research.
D-Chiro-Inositol	50 mg twice daily	With Myo-Inositol (40:1 ratio)	Taken with myo-inositol in the physiological 40:1 ratio. DCI alone in high doses worsens oocyte quality — always pair with myo. Many quality supplements contain both in correct ratio.
Magnesium Glycinate	300–400 mg	With dinner or 1 hour before bed	Magnesium is a cofactor in over 300 enzymatic reactions including insulin receptor phosphorylation. Studies show 80%+ of PCOS women are deficient. Also improves sleep quality and cortisol clearance.
Vitamin D3 + K2	5,000 IU D3 · 100 mcg K2 MK-7	With largest meal (fat-soluble)	Vitamin D deficiency is nearly universal in PCOS and inversely correlated with insulin resistance severity. K2 ensures calcium routing — do not supplement D3 without K2 long-term. Dose based on

Supplement	Dose	Timing	Why
			labs if available.
Omega-3 (EPA + DHA)	2–3 g combined EPA/DHA daily	With meals (split dose if GI-sensitive)	Reduces VLDL triglycerides, inflammatory cytokines, and free androgen index. High-quality fish oil — look for IFOS-certified brands. Plant-based ALA does not substitute; conversion to EPA/DHA is insufficient in IR states.
Zinc Bisglycinate	25–30 mg daily	With food (evening preferred)	Zinc inhibits 5-alpha-reductase (converts testosterone to DHT — the potent form driving acne and hair loss). Studies show zinc supplementation reduces free testosterone and improves ovulation rates. Take with food to avoid nausea.

📋 Week 1 Workbook

Starting Measurements (Day 1)

Marker	Day 1 Value	Day 20 Value	Day 60 Value
Fasting glucose (mg/dL or mmol/L)			
Fasting insulin (µIU/mL)			
HOMA-IR			
Free testosterone			
SHBG			
hs-CRP			
Vitamin D (25-OH)			
Weight / waist measurement			

Marker	Day 1 Value	Day 20 Value	Day 60 Value
Last period date			

Week 1 Daily Tracker

Habit	D1	D2	D3	D4	D5	D6	D7
Protein at every meal ✓							
No sugar / refined carbs							
Post-meal walk (min)							
Myo-Inositol AM + PM							
Magnesium (evening)							
Omega-3 + D3							
Sleep duration (hrs)							
Caffeine cutoff met							

Reflection Prompts — End of Week 1

- What was the hardest food to remove? What did you reach for instead?
- Describe your energy at 3pm compared to your baseline before the protocol.
- Did you notice any difference in hunger patterns with structured meal timing?

Week Two · Days 8–14

Pattern Recognition

Your blood sugar tells you what your labs can't — learn to read it

You have removed the worst offenders. Now you start observing how your body responds to what remains. This week is about data collection, not restriction.

Using Your CGM (or Glucose Meter) This Week

What to Measure	Target Range	What It Tells You
Fasting glucose (on waking)	70–90 mg/dL (3.9–5.0 mmol/L)	Your insulin sensitivity overnight; elevated fasting glucose = impaired insulin clearance
1-hour post-meal peak	<140 mg/dL (<7.8 mmol/L)	How that meal triggered insulin; anything above 140 is a metabolic stress event
2-hour post-meal	Back to baseline (<120 mg/dL)	How efficiently you clear glucose; a slow return = significant IR
Pre-bedtime glucose	80–100 mg/dL	Going to bed hyperglycemic worsens morning insulin resistance (the dawn phenomenon)

The PCOS Stress Spike

Check your fasting glucose after a stressful morning even before eating. Women with PCOS often show elevated fasting glucose driven purely by cortisol — the liver dumps glycogen in response to morning cortisol rise (dawn phenomenon, amplified by HPA dysregulation). If your fasting glucose is regularly above 95 without eating, cortisol is a significant variable and Phase 3 becomes critical for you.

Nutrition: Week 2 Additions

Maintain Week 1 removals. Add these evidence-backed strategies:

- Vinegar before high-carb meals: 1–2 tsp of raw apple cider vinegar in water 15–20 minutes before any meal containing starchy vegetables or legumes. Acetic acid inhibits alpha-amylase, slowing glucose absorption. Published data shows up to 20–30% reduction in post-meal glucose spike.
- Resistant starch: Cooked-and-cooled rice, cooked-and-cooled potatoes, and green plantains are significantly lower glycemic than their fresh-cooked equivalents. If you are including starch, preparation matters.
- Spearmint tea — 2 cups daily: Spearmint has demonstrated anti-androgenic activity in clinical trials, reducing free testosterone levels in PCOS without affecting other androgens. Consistent daily use — not occasional. Use as morning and afternoon beverage.
- Minimum 30g fiber daily: Soluble fiber (chia seeds, flaxseed, psyllium husk) slows gastric emptying and blunts post-meal glucose. Insoluble fiber (leafy greens, broccoli, celery) feeds estrogen-clearing gut bacteria. Both matter.

Movement: Week 2 — Add Resistance Training

Skeletal muscle is the primary site of insulin-mediated glucose disposal. More muscle = better glucose uptake. This is non-negotiable for long-term insulin sensitivity reversal.

Resistance Training Protocol

Frequency: 2–3 sessions this week, 30–40 min each Format: Compound movements — squats, deadlifts, rows, presses. No isolation machines as primary work. Intensity: Challenging but not to failure. You should be able to complete each set with 1–2 reps left in reserve. Timing: Before meals when possible — pre-meal resistance training sensitizes muscle to insulin for 24–48 hours post-session. Recovery: 48 hours between sessions.

> Sleep is where the adaptation happens.

📋 Week 2 Workbook

CGM / Glucose Log

Reading	D8	D9	D10	D11	D12	D13	D14
Fasting (mg/dL)							
1hr post-Meal 1							
1hr post-Meal 2							
Pre-bed (mg/dL)							
Resistance training ✓							
Post-meal walks ✓							
Spearmint tea (2 cups)							

Identify Your Personal Glucose Triggers

List the top 3 meals or foods that caused your highest post-meal glucose spike this week.

This is your personal data — not a judgment.

Week 2 Reflection

- What surprised you about your glucose response this week?
- How did your energy and hunger differ on days you did resistance training vs. days you didn't?
- What symptoms (skin, mood, cravings, cycle-related) did you notice changing — if any?

Week Three · Days 15–20

Foundation Lock-In

Consolidate, assess, and prepare for Phase 2

Add Berberine — The Phase 1 Closer

Berberine is the supplement with the most robust head-to-head comparison to metformin in insulin resistance research. It activates AMPK (the same pathway as metformin), improving glucose uptake, reducing hepatic glucose output, and lowering fasting insulin. For women who cannot tolerate or access metformin — or who want a non-prescription alternative — berberine is the closest evidence-based proxy available.

⚠️ Berberine — Critical Notes

- Do not take berberine if you are pregnant or trying to conceive immediately — animal studies show uterotonic effects. Discontinue if actively pursuing conception.
- If you are currently on metformin, do not add berberine without physician consultation — combined use can cause hypoglycemia.
- Berberine reduces the absorption of some medications — take 2 hours away from any prescription drugs.
- Start at 500mg once daily with a meal for 3 days before increasing to the full dose to reduce GI side effects.

Supplement	Dose	Timing	Why
Berberine HCl	500 mg, 2–3× daily	With each main meal (introduce gradually)	AMPK activator. Reduces fasting glucose, lowers HbA1c, improves lipid profile. Multiple meta-analyses demonstrate equivalence to metformin for insulin

Supplement	Dose	Timing	Why
			sensitization in PCOS populations. Use berberine phytosome for higher bioavailability.
N-Acetyl Cysteine (NAC)	600 mg, 2× daily	Morning and afternoon, with food	NAC is the precursor to glutathione — your primary antioxidant. In PCOS, oxidative stress impairs insulin receptor function and drives ovarian androgen production. Studies show NAC reduces free testosterone, improves insulin sensitivity, and supports ovulation — comparable to clomiphene in some fertility trials.

Phase 1 Assessment — Day 20

Before entering Phase 2, verify you have established these non-negotiables:

- Fasting glucose <100 mg/dL on most mornings
- Post-meal spikes consistently under 140 mg/dL
- Protein at every meal, no skipped meals
- Post-meal walks happening at least once daily
- Resistance training 2–3× per week
- Full supplement stack from Week 1 in routine
- Berberine introduced and tolerated
- Sleep averaging 7+ hours

If fewer than 6 of these are true: spend Days 20–25 reinforcing Phase 1 before advancing.

Phase 2 interventions work on top of this foundation, not instead of it.

📋 Phase 1 Close — Day 20 Check-In

Symptom Shift Log

Symptom	Before Protocol (1–10)	Day 20 (1–10)	Notes
Energy crashes (afternoon)			
Sugar / carb cravings			
Skin (acne severity)			
Bloating / digestive issues			
Mood stability			
Sleep quality			
Anxiety / irritability			
Hair shedding			
Hirsutism (perceived)			

What Has Been Hardest About Phase 1?

What Has Surprised You Most?

Hormonal Recalibration.

Week Four · Days 21–27

Inflammation Reset

Insulin resistance and inflammation are a closed feedback loop — break it from both ends

Chronic low-grade inflammation is both a consequence and a driver of insulin resistance in PCOS. Elevated IL-6, TNF-α, and CRP independently impair insulin receptor sensitivity. This week adds anti-inflammatory nutrition and liver support.

Anti-Inflammatory Nutrition Upgrade

Add This Week	Mechanism	Practical Form
Turmeric + black pepper	Curcumin inhibits NF-κB and COX-2; black pepper (piperine) increases curcumin bioavailability by 2,000%	1/2 tsp turmeric + pinch black pepper in cooking daily, or supplement form
Fatty fish 3–4× weekly	EPA/DHA reduce prostaglandin E2, IL-1β, and directly compete with arachidonic acid in inflammatory pathways	Salmon, sardines, mackerel, anchovies — wild-caught preferred
Cruciferous vegetables daily	Indole-3-carbinol and DIM support estrogen metabolism and clear excess androgens via liver Phase 2 detoxification	Broccoli, cauliflower, Brussels sprouts, kale — lightly steamed preserves most I3C content
Ground flaxseed (2 tbsp daily)	Lignans compete with androgens at receptor sites; soluble fiber feeds SCFA-producing gut bacteria that regulate estrogen clearance	Add to smoothies, yogurt, salads — must be ground, not whole, for absorption
Reduce caffeine to 1 cup before 10am	Caffeine elevates cortisol, which drives adrenal DHEA-S and inflammatory arachidonic acid production; also impairs iron absorption and sleep	Substitute: spearmint tea, rooibos, ginger-lemon water in afternoons

Gut Health — The Underestimated PCOS Variable

Research published in the last decade has confirmed significant gut microbiome dysbiosis in PCOS populations — reduced Lactobacillus species, elevated endotoxin-producing bacteria (LPS), and impaired short-chain fatty acid (SCFA) production. These directly worsen insulin resistance and increase inflammatory load.

Gut Protocol Additions — Week 4

- Probiotic: Multi-strain formula containing Lactobacillus acidophilus, L. rhamnosus, Bifidobacterium longum . Take on an empty stomach in the morning. Look for minimum 20 billion CFU with documented strains (not proprietary blends).
- Prebiotic fiber: 5–10g psyllium husk in water before the largest meal. Feeds the SCFA-producing bacteria depleted in PCOS.
- Fermented foods — 1 serving daily: Plain full-fat kefir (if dairy-tolerant), kimchi, sauerkraut, or water kefir. Start small — 2 tbsp — and increase gradually to avoid bloating.

📋 Week 4 Workbook — Inflammation Tracking

Inflammatory Food Log

Note any foods that seem to trigger bloating, brain fog, skin flares, or joint stiffness within 2–4 hours. This is your personalized inflammatory trigger data.

Week 4 Habit Tracker

Habit	D21	D22	D23	D24	D25	D26	D27
Cruciferous veg							
Ground flaxseed							
Probiotic (AM)							
Spearmint tea ×2							
Resistance training							
Sleep ≥7 hrs							

Week Five · Days 28–34

Androgen Regulation

Lower free testosterone at the source and at the receptor

By Week 5, insulin should be trending down and inflammation lower. This creates the environment for targeted androgen-reducing strategies to work. Adding these before Phase 1 is established is largely ineffective — the elevated insulin will override them.

Androgen Reduction — The Three Levers

Lever 1: Reduce Production

Lower insulin (Phase 1 work) reduces ovarian stimulus. NAC and myo-inositol also directly reduce theca cell androgen synthesis. Zinc inhibits the androgen-amplifying enzyme 17β-HSD.

Lever 2: Increase Binding

Raise SHBG so more testosterone is bound and inactive. SHBG rises as insulin falls. Cruciferous vegetables, flaxseed, and weight distribution toward muscle all increase SHBG production.

Lever 3: Block Conversion

5-alpha-reductase converts testosterone to DHT — the androgen responsible for scalp hair loss and facial hair. Zinc, saw palmetto (160mg standardized), and green tea extract inhibit 5-alpha-reductase. Spearmint blocks androgen receptors directly. Reishi mushroom shows early evidence for 5-alpha-reductase inhibition.

Add to Supplement Stack — Week 5

Supplement	Dose	Timing	Why
Saw Palmetto	160 mg standardized extract	With dinner	5-alpha-reductase inhibitor. Reduces DHT conversion without blocking testosterone production entirely. Evidence primarily from male pattern baldness research but mechanism applies directly to PCOS-related hirsutism and androgenic alopecia. Takes 8–12 weeks to show hair-related changes.
Licorice Root Extract	150–300 mg glycyrrhizic acid standardized	With lunch — cycle use: 3 weeks on, 1 week off	Clinical trials show licorice root reduces testosterone in women with PCOS by inhibiting 17β-HSD (testosterone synthesis enzyme). Do NOT use if you have hypertension — glycyrrhizin raises blood pressure. Do not use for more than 3 continuous weeks without a break. Use with caution and monitor BP.

Cycle Tracking Begins Here

If you are not currently tracking your cycle, start now. You cannot measure hormonal progress without baseline data. Use a paper chart, an app (Clue, Natural Cycles, Tempdrop), or a basal body temperature (BBT) thermometer. Record:

- Cycle day (Day 1 = first day of full menstrual flow)
- BBT daily (morning, before rising, same time each day)
- Cervical mucus quality (dry, creamy, stretchy/egg-white)
- Any bleeding, spotting, or cycle-related symptoms

- Energy and mood by cycle phase

📋 Week 5 Workbook — Androgen & Cycle Tracking

Current Cycle Status

Question	Your Answer
Last period start date	
Average cycle length (or "irregular")	
Do you detect ovulation signs?	
Primary androgen symptoms	
Most bothersome symptom to resolve	

Androgen Symptom Rating — Week 5

Symptom	Day 1 Rating (1–10)	Week 5 Rating (1–10)	Change
Acne (face/back/chest)			
Facial / body hair growth			
Scalp hair thinning			
Oily skin			
Mood / irritability			

Week 5 Reflection

- Have you noticed any changes in skin, hair, or cycle regularity since Day 1?
- What is your biggest remaining symptom burden?

Week Six · Days 35–40

Mid-Protocol Assessment

Measure, adjust, and prepare for the final phase

Retest Labs at Day 40

If you ran baseline labs, this is the time to retest the key markers. Even without labs, your symptom tracking and glucose data provide meaningful signal.

Marker	Baseline	Day 40	Target Direction
Fasting insulin			↓ Significant drop expected
HOMA-IR			↓ <2.0 by Day 40 is achievable
Free testosterone			↓ Typically begins falling by Week 4–5
SHBG			↑ Should be rising with lower insulin
hs-CRP			↓ Inflammation reduction from Phase 2 work
Fasting glucose			↓ Should be reliably under 90 mg/dL

If Results Are Minimal at Day 40

Check for: Sleep quality issues (address in Phase 3). Hidden glucose spikes from foods you consider "safe." Inadequate protein intake (common underestimation — track for 3 days). Undiagnosed thyroid dysfunction (TSH above 2.5 will blunt insulin sensitivity improvement). Significant ongoing stress load (cortisol overrides most of this protocol). These are not failure points — they are diagnostic signals.

📋 Phase 2 Close — Midpoint Review

What's Working — Be Specific

What's Not Working or Unclear

Protocol Adjustments I'm Making Going Into Phase 3

Metabolic Resilience.

Week Seven · Days 41–47

Advanced Insulin Strategies

Layer precision tools onto an already-stable foundation

Introduce Time-Restricted Eating (Now)

You have 40 days of stable blood sugar as a foundation. Now time-restricted eating (TRE) can be introduced safely without triggering the cortisol response that made it counterproductive in Phase 1.

> **Protocol: 14:10 Window**
>
> Eating window: 8am – 6pm (or 9am – 7pm — whichever fits your lifestyle) Fasting window: 14 hours overnight Not a calorie restriction. You eat the same food volume inside the window. If you experience significant hunger, dizziness, or increased anxiety: Your cortisol response is activated — widen the window back to 12 hours and do not force it. TRE works best when it feels natural, not when it's a fight. Do not use TRE on resistance training days if your training falls outside the eating window — protein synthesis requires adequate fueling around the session.

Cycle-Synced Nutrition

If you are experiencing any cycle regularity (even irregular cycles), your nutritional needs shift across the four phases. Insulin sensitivity is naturally higher in the follicular phase (days 1–14) and lower in the luteal phase (days 15–28 of a regular cycle).

Cycle Phase	Hormonal Context	Nutritional Emphasis	Movement Emphasis
Menstrual	Low estrogen, low	Iron-rich foods, anti-inflammatory	Gentle — walking,

Cycle Phase	Hormonal Context	Nutritional Emphasis	Movement Emphasis
(Day 1–5)	progesterone, high prostaglandins	emphasis, adequate calories — do not restrict during menstruation	yoga, stretching
Follicular (Day 6–13)	Rising estrogen → peak insulin sensitivity	Higher carbohydrate tolerance window; best time for dietary experimentation; fermented foods	Higher intensity — HIIT if desired, heavy resistance
Ovulation (Day 14–16)	Estrogen peak, LH surge	Zinc-rich foods support egg quality; cruciferous vegetables for estrogen clearance post-ovulation	High output — peak strength, peak endurance
Luteal (Day 17–28)	Progesterone rises, insulin sensitivity drops 25–30%	Lower carbohydrates, higher fat, higher magnesium, B6, zinc; cravings are real — do not ignore them, redirect them	Moderate — strength at lower intensity, longer walks, pilates

For Women Without Detectable Cycles

If cycles are absent (amenorrhea) or completely irregular, use a 28-day rotating template — treat days 1–14 as follicular and days 15–28 as luteal. This is not biologically precise, but it approximates the metabolic variation that cycle syncing addresses. As your cycle regularizes over months on this protocol, adjust to your actual pattern.

📋 Week 7 Workbook

TRE Log

Day	D41	D42	D43	D44	D45	D46	D47
First meal time							
Last meal time							

Day	D41	D42	D43	D44	D45	D46	D47
Fasting window (hrs)							
Hunger/anxiety (1–5)							
Fasting glucose AM							

Week Eight · Days 48–54

The Cortisol-PCOS Loop

The stress hormone runs a parallel androgen production line — close it

> "You can eat perfectly and still spike androgens if your cortisol is chronically elevated. Stress is not a mental health problem. It is a metabolic one."

How Cortisol Drives PCOS Independently

The adrenal glands produce androgens independently of the ovaries — primarily DHEA-S, which converts to testosterone. Cortisol and DHEA-S are co-produced by the adrenal cortex. Chronic stress (HPA axis hyperactivation) elevates both. This is why women under significant psychological, physical, or metabolic stress see worsening PCOS symptoms regardless of diet — the adrenal pathway is running in parallel.

Cortisol Driver	PCOS Consequence	Intervention
Inadequate sleep (<7 hours)	Morning cortisol spike → elevated fasting glucose and DHEA-S; reduces next-day insulin sensitivity by 20–25%	Sleep architecture optimization (below)
Chronic under-eating / calorie restriction	Metabolic starvation signal → cortisol elevation → adrenal androgen production; explains why aggressive calorie cutting worsens PCOS	Adequate protein and caloric intake; no extended fasting beyond 16 hours
Over-exercising / HIIT excess	High-intensity exercise raises cortisol acutely; chronic HIIT without adequate recovery keeps it elevated; amplifies inflammatory load	2–3 resistance sessions + walking; HIIT maximum 1× weekly, only in follicular phase
Chronic psychological	Sustained HPA activation; cortisol-	Structured stress regulation

Cortisol Driver	PCOS Consequence	Intervention
stress	induced inflammation; impaired sleep → compounding IR effect	protocols (below)
Blood sugar instability	Hypoglycemia is a physiological stressor — cortisol releases to restore glucose; unstable blood sugar = sustained cortisol elevation	Phase 1 work addresses this directly; consistent meal timing and protein intake are cortisol-regulating tools

Sleep Architecture — The Non-Negotiable

Insulin sensitivity is restored during deep sleep via growth hormone pulses and cortisol clearance. Women with PCOS show measurably poorer sleep architecture independent of body weight — elevated night-time cortisol, shorter deep sleep phases, and higher rates of sleep apnea (even without obesity).

Sleep Optimization Protocol

- Fixed sleep/wake times: Within 30 minutes, 7 days a week. Circadian rhythm stability directly impacts cortisol peak timing and insulin sensitivity restoration.
- Target: 7.5–9 hours in bed. Not optional. Not a luxury. Non-negotiable for metabolic restoration.
- Light exposure protocol: 10+ minutes of natural light within 30 minutes of waking (sets circadian clock); block all screens 90 minutes before bed or use blue-light blocking glasses.
- Temperature: Room temperature 16–19°C (60–67°F) for optimal deep sleep.
- Evening carbohydrate strategy: A small serving of complex carbohydrates (50–100 calories) at the evening meal supports serotonin → melatonin conversion. Do not eliminate carbohydrates entirely from the evening meal if sleep is poor.
- Magnesium glycinate (already in your stack) at 300–400mg 30–60 min before bed supports GABA receptor activity and deep sleep onset.

Add to Supplement Stack — Week 8

Supplement	Dose	Timing	Why
Ashwagandha (KSM-66)	300–600 mg	Evening, with dinner	Adaptogenic herb with the highest evidence base for cortisol reduction. KSM-66 extract reduces serum cortisol by up to 27% in published trials, improves thyroid markers, and shows improvements in DHEA-S in stressed women. Do not use if you have Hashimoto's thyroiditis without physician guidance — nightshade-adjacent compounds may trigger immune response in susceptible individuals.
Phosphatidylserine	200–400 mg	With evening meal or 1 hour before bed	Phospholipid that blunts post-exercise and post-stress cortisol elevation. Particularly useful for women who exercise intensely or have chronically elevated evening cortisol (confirmed with DUTCH test or 4-point salivary cortisol panel). Supports sleep quality and cognitive function under stress.
B6 (Pyridoxal-5-Phosphate, P5P)	50 mg	With dinner	P5P is the active form of B6 — required for progesterone synthesis, serotonin conversion, and cortisol regulation. Deficiency amplifies PMS-type symptoms in the luteal phase and impairs insulin metabolism. Avoid high-dose B6 (above 100mg) long-term without lab guidance — peripheral neuropathy risk

Supplement	Dose	Timing	Why
			with excess.
L-Theanine	200 mg	30–60 minutes before bed	Non-sedating anxiolytic amino acid from green tea. Promotes alpha-wave brain activity, reduces anxiety without sedation, and improves sleep onset without dependence. Safe for long-term use. Can also be used during the day for stress-driven glucose spikes.

Structured Stress Regulation

The goal is not to eliminate stress. It is to reduce the duration of cortisol elevation after a stressor. The following techniques have published evidence for HPA axis downregulation:

- Physiological sigh (twice daily): Double inhale through the nose (filling lungs fully, then sipping in extra air) followed by a long slow exhale through the mouth. One to two repetitions. This is the fastest-acting, most evidence-supported cortisol downregulation tool available without a prescription.
- 4-7-8 breathing before meals: Activates the parasympathetic nervous system, improving vagal tone. Better vagal tone = improved gut motility and insulin sensitivity during the meal.
- Deliberate cold exposure (optional): 30–60 second cold shower at the end of your normal shower. Acute cold stress followed by recovery builds HPA resilience. Do not attempt during menstruation or on days of significant fatigue.
- Minimum 20 minutes of non-goal-directed outdoor time daily. Unstructured nature exposure reduces salivary cortisol in controlled studies. Not a walk with a fitness goal — walking without your phone, without a route objective.

📋 Week 8 Workbook — Cortisol & Sleep

Sleep Tracking

Day	D48	D49	D50	D51	D52	D53	D54
Bedtime							
Wake time							
Sleep quality (1–5)							
Fasting glucose AM							
Stress level (1–5)							
Ashwagandha taken							
Breathing practice							

Stress-Symptom Connection

- Describe your highest-stress day this week. What were your glucose readings and symptoms that day?
- What cortisol driver from the table above is most active in your life right now?

Final Days · Days 55–60

Integration & What Comes Next

The protocol ends. The practice continues.

Final Labs — Day 60

Request a repeat of your baseline panel. The changes you are looking for:

Marker	Realistic 60-Day Target	What Further Progress Looks Like
Fasting insulin	30–50% reduction from baseline	Continued decline over 6–12 months with maintained habits
HOMA-IR	<2.0 if baseline was >3.0; <1.5 if baseline was 2.0–3.0	Optimal: <1.0 — achievable at 6 months
Free testosterone	Measurable decrease; may not normalize in 60 days	Full normalization typically requires 3–6 months
SHBG	Clear upward movement	Continues rising for months as insulin falls
hs-CRP	<1.0 mg/L	Optimal: <0.5 mg/L
Cycle regularity	Many women see first ovulation signal within 60–90 days	Consistent ovulation typically returns at 3–6 months of maintained protocol

Your Long-Term Protocol Framework

Keep Indefinitely (Non-Negotiable)

- Protein at every meal
- Post-meal walks (at least after dinner)
- Myo-inositol + D-chiro-inositol (maintenance: 2g/day total)
- Magnesium glycinate
- Omega-3

- Resistance training 2–3× weekly
- Sleep ≥7.5 hours
- Spearmint tea daily

Cycle or Reassess Every 3 Months

- Berberine (can reduce to 500mg once daily if HOMA-IR <1.5)
- Vitamin D (retest and adjust dose)
- Licorice root (use 3 weeks on, 1 week off cycle)
- Saw palmetto (reassess need at 6 months based on androgens)
- NAC (continue 6 months minimum, then assess)
- Berberine — cycle 8 weeks on, 2 weeks off for long-term use

What Will Undermine Everything You've Built

The Relapse Triggers — Know Them In Advance

- Returning to "just a little" sugar or refined carbs as a daily habit. For women with insulin resistance, maintenance requires maintaining the structure — not perfection, but consistent structure. Occasional exceptions are metabolically tolerable; daily exceptions are not.
- Extended sleep deprivation (travel, work sprints, newborns). Even 3–4 nights of poor sleep can measurably reverse insulin sensitivity gains. Acknowledge it, protect sleep as aggressively as possible, and reestablish as quickly as you can.
- Eliminating all carbohydrates long-term without medical supervision. Severe carbohydrate restriction long-term in women with HPA axis vulnerability can suppress thyroid function and worsen cortisol dysregulation. This protocol uses low-carb to stabilize, not zero-carb to maintain permanently.

- Stopping supplements and expecting food alone to maintain the gains. The supplement stack addresses deficiencies and provides pharmacological support that food cannot fully replicate for most PCOS phenotypes. Maintain the core stack.

📋 Day 60 Final Review

Final Symptom Assessment

Symptom	Day 1 (1–10)	Day 60 (1–10)	Your Notes
Energy / afternoon crash			
Sugar cravings			
Acne severity			
Hirsutism / hair growth			
Scalp hair loss			
Cycle regularity			
Sleep quality			
Mood / anxiety			
Bloating / digestion			
Overall energy level			

60-Day Protocol Reflection

- What is the single most significant change you observed — in labs, symptoms, or how you feel?
- What was the hardest part of this protocol, and what would you do differently?
- What are you most committed to maintaining beyond Day 60?

- What question do you still have that this protocol didn't fully answer?

Master Supplement Reference

All supplements referenced in this protocol, organized by introduction phase with full rationale and contraindications.

Supplement	Phase	Dose	Timing	Key Contraindications
Myo-Inositol	Phase 1, Week 1	2,000 mg ×2 daily	AM + PM with meals	None known at physiological doses

Supplement	Phase	Dose	Timing	Key Contraindications
D-Chiro-Inositol	Phase 1, Week 1	50 mg ×2 daily	With myo-inositol	Do not use in high doses without myo ratio
Magnesium Glycinate	Phase 1, Week 1	300–400 mg	Evening / before bed	Reduce dose if loose stools occur
Vitamin D3 + K2	Phase 1, Week 1	5,000 IU D3 / 100 mcg K2	With largest meal	Adjust dose to labs; hypercalcemia risk with excess
Omega-3 (EPA/DHA)	Phase 1, Week 1	2–3 g combined	With meals	Blood thinners — physician consultation required
Zinc Bisglycinate	Phase 1, Week 1	25–30 mg	With evening meal	Take with food; copper depletion with very long-term use
Berberine HCl	Phase 1, Week 3	500 mg ×2–3 daily	With meals	Not with metformin; not in pregnancy; 2hr gap from medications
NAC	Phase 1, Week 3	600 mg ×2 daily	Morning + afternoon	Avoid with nitroglycerin; GI sensitivity — take with food
Probiotic	Phase 2, Week 4	≥20 billion CFU multi-strain	Empty stomach, AM	SIBO risk — consult physician if GI symptoms worsen
Saw Palmetto	Phase 2, Week 5	160 mg standardized	With dinner	Rare: GI upset; theoretical hormonal interactions — discuss with prescribing physician
Licorice Root	Phase	150–300 mg	With lunch; 3	Contraindicated in

Supplement	Phase	Dose	Timing	Key Contraindications
	2, Week 5	glycyrrhizic acid	weeks on, 1 off	hypertension, kidney disease, pregnancy, heart failure
Ashwagandha (KSM-66)	Phase 3, Week 8	300–600 mg	Evening	Caution with Hashimoto's; not in pregnancy; avoid with sedative medications
Phosphatidylserine	Phase 3, Week 8	200–400 mg	Evening with meal	Blood thinners — use caution; start at lower dose
B6 (P5P form)	Phase 3, Week 8	50 mg	With dinner	Do not exceed 100mg daily long-term without monitoring
L-Theanine	Phase 3, Week 8	200 mg	Before bed	None known; can combine with magnesium

Biomarkers: What to Track and What the Numbers Mean

Marker	Optimal (Functional)	Flag (Investigate)	PCOS Relevance
Fasting insulin	<5 µIU/mL	>10 µIU/mL	Primary driver of androgen overproduction; most sensitive early indicator
HOMA-IR	<1.5	>2.0	Composite insulin resistance score; calculated, not tested directly
Fasting glucose	70–85 mg/dL	>95 mg/dL	"Normal" range (under 100) can mask significant insulin resistance
HbA1c	<5.3%	>5.6%	3-month glucose average; misses post-meal spikes

Marker	Optimal (Functional)	Flag (Investigate)	PCOS Relevance
Free testosterone	0.3–1.9 ng/dL (lab-dependent)	Above reference range	Direct androgen burden; most relevant for acne, hirsutism, hair loss
SHBG	40–120 nmol/L	<30 nmol/L	Low SHBG = high free androgen availability regardless of total testosterone
LH:FSH ratio	<2:1	>2:1 (or LH >FSH)	Elevated LH relative to FSH impairs follicle maturation and ovulation
DHEA-S	Age-appropriate range	Above upper quartile for age	Adrenal androgen marker; elevated = significant adrenal contribution to PCOS
hs-CRP	<0.5 mg/L	>1.0 mg/L	Inflammatory burden; amplifies insulin receptor dysfunction
Vitamin D (25-OH)	50–80 ng/mL	<30 ng/mL	Deficiency strongly correlated with IR severity and androgen excess in PCOS
Ferritin	50–100 ng/mL	<30 or >150 ng/mL	Low ferritin impairs mitochondrial energy metabolism; high ferritin = inflammation marker
TSH	0.5–2.5 mIU/L	>2.5 mIU/L	Subclinical hypothyroidism blunts insulin sensitivity improvement; address first if elevated

LIV Functional

PCOS does not have to be managed indefinitely. For most women, it is a reversible metabolic state — one that responds directly to the removal of its root driver: chronic hyperinsulinemia.

> "Sixty days is not enough to undo a decade. It is enough to prove your biology can change

> — and to build the habits that make it permanent."

Continue with LIV Functional

This protocol gives you the framework. What accelerates results is personalisation: your CGM patterns, your cortisol profile, your thyroid and hormonal status, and the specific inputs your biology responds to.

Ready for a deep dive session with Reham?

A diagnostic session to find what is actually blocking your numbers, and to build your plan on your data rather than on population averages. Or start coaching with me directly.

Book your session | **Get coached**

— Reham · **Liv Functional**

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